Parardha Dhar

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github.com/parardhadhar | linkedin.com/in/parardhadhar | parardha.vercel.app

PROFILE

Computer Science undergraduate at VIT Bhopal with research internship experience at IISc Bangalore. Built and deployed full-stack applications, webhook automation systems, and AI integrations using TypeScript, Python, PostgreSQL, and Redis, including a community platform serving 1,800+ unique visitors.

EDUCATION

VIT Bhopal University | B.Tech., Computer Science and Engineering

Sep 2024 - Expected 2028 | **GPA: 8.45/10.0**

Coursework: Data Structures and Algorithms, Operating Systems, Database Management Systems, Machine Learning

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL

Backend & data: Node.js, Express, FastAPI, REST APIs, PostgreSQL, Redis, BullMQ, SQLAlchemy

Frontend: React, Next.js, Tailwind CSS, Electron | **AI/ML:** PyTorch, OpenCV, scikit-learn, Groq, LLM APIs

Tools: Git, Docker, GitHub Actions, Linux/Unix, Vercel, Supabase, Firebase

EXPERIENCE

Machine Learning Research Intern | IISc Bangalore | June - July 2026

Computational Materials Engineering | TEM AI Studio | Source code

- Built electron-microscopy analysis software, reducing manual defect calculation from **2+ hours to under 15 seconds per micrograph**.
- Fine-tuned PyTorch denoising diffusion models for noisy lattice reconstruction, improving downstream YOLOv11 defect detection accuracy by **45%**.
- Integrated FastAPI with a React/Electron interface for model inference; implemented rate-limited endpoints and streaming JSON responses for an LLM copilot.
- Implemented OpenCV-based 2D FFT analysis and geometric Burgers vector calculations for **50+ megapixel** scans.

SELECTED PROJECTS

FlowHook | AI-Assisted Webhook Automation Engine

github.com/parardhadhar/FlowHookCore - Code, screenshots & local demo

TypeScript, Node.js, Express, PostgreSQL, Redis/BullMQ, Next.js, Groq, Docker

- Built signed GitHub/Stripe webhook ingestion with per-workflow duplicate detection, background workers, five-attempt exponential retries, and persisted execution traces.
- Integrated Groq issue summarization into email/Slack workflows with bounded inputs, timeouts, deterministic fallbacks, and cached summaries across action retries.
- Verified signed duplicate delivery, worker execution, and captured email in Docker; passed 13 signature and AI tests.

Grahak | AI-Assisted WhatsApp CRM

grahak-flame.vercel.app (early access) | github.com/parardhadhar/grahak

Next.js, TypeScript, FastAPI, SQLAlchemy, JWT, Groq, Baileys

- Built a customer inbox, searchable contacts, notes, and follow-ups with JWT authentication and business-scoped data access; verified backend behavior with **23 tests**.
- Integrated Groq conversation summaries and customer-detail extraction with provider timeouts and fallbacks; implemented WhatsApp sync with duplicate protection and a persistent retry outbox.

VRGC Community Hub | Full-Stack Platform | vrgc.live

Next.js, React, Supabase, Firebase, Vercel, CI/CD

- Independently built and deployed a university community platform serving **1,800+ unique visitors and 8,300+ page views**; managed live operations during a high-traffic event.
- Achieved a **96/100 Vercel Real Experience Score**, with 1.32-second FCP and 1.48-second LCP.

LEADERSHIP

Founder & President | Virtual Reality and Gaming Club (VRGC), VIT Bhopal

CERTIFICATIONS & PROBLEM SOLVING

Oracle: OCI 2025 Generative AI Professional | **ISRO:** AI/ML for Geo Data Analysis

DeepLearning.AI: Supervised Machine Learning | **IBM:** Machine Learning with Python

Solved **100+ programming problems** across LeetCode, Codeforces, GeeksforGeeks, and HackerRank.